Lab 1: Requirements Engineering & Traceability

Problem Statement #33: Micro-Lending & Peer Credit Risk Assessor

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1. Problem Context & Overview

The **Micro-Lending & Peer Credit Risk Assessor** platform is a decentralized peer-to-peer micro-finance ecosystem engineered to expand responsible credit access to underserved borrowers. It calculates deterministic borrower credit risk scores from debt-to-income (DTI) metrics, produces exact reducing-balance repayment schedules, orchestrates fractional peer investments, and triggers automated default warnings upon missed deadlines.

2. Functional Requirements Table (FR-001 to FR-005)

ID	Priority	Description	Acceptance Criteria	Rationale
FR-001	High	Compute borrower credit risk ratings and numerical scores (300–850) based on DTI, repayment history, and employment stability.	**Pass:** Risk tier (Low/Med/High) accurately assigned. **Fail:** Incomplete profile generates approval.	Automates underwriting without bias while safeguarding peer lender capital.
FR-002	High	Generate structured reducing-balance EMI schedules with detailed principal/interest splits upon loan approval.	**Pass:** Sum of principal matches sanctioned amount (diff ≤ Rs.0.50). **Fail:** Negative amortization or rounding drift.	Provides transparent repayment obligations for borrowers and cash flow projections for lenders.
FR-003	High	Enable Peer Lenders to browse active loans, filter by risk tiers, and commit fractional funds into escrow.	**Pass:** Escrow locked atomically, listing marked Fully Funded at 100%. **Fail:** Overfunding permitted or wallet deducted without lock.	Enables fractional risk diversification across multiple micro-borrowers.
FR-004	Medium	Process borrower installment payments via gateway and distribute proportional dividends to participating lenders.	**Pass:** Proportional shares credited inside an ACID transaction. **Fail:** Borrower debited but lender wallets unchanged.	Preserves financial ledger integrity and guarantees fair dividend returns.
FR-005	High	Monitor repayment due dates and automatically trigger tiered default warnings when grace periods expire.	**Pass:** T+3 days triggers reminder; T+15 days triggers formal default. **Fail:** Warning fires on on-time accounts or fails to fire.	Enforces disciplined repayment and minimizes platform Non-Performing Assets.

3. Non-Functional Requirements (NFR-001 & NFR-002)

ID & Type	Priority	Description	Acceptance Criteria	Rationale
NFR-001 Performance & Security	High	Repayment transactions, escrow locks, and lender dividend payouts must execute inside ACID transactions with strict serializability.	**Pass:** 200 concurrent requests verify zero ledger discrepancies, 100% atomic rollbacks, latency < 350 ms. **Fail:** Race conditions or double debits.	Ledger consistency is mission-critical; financial inconsistencies create legal liability and violate RBI P2P mandates.
NFR-002 Reliability & Privacy	High	Borrower PII, tax IDs, and credit data must be encrypted at rest (AES-256) and in transit (TLS 1.3), with backups ensuring RPO ≤ 15m, RTO ≤ 1h.	**Pass:** Zero plaintext PII in database, TLS 1.3 enforced, disaster recovery restores operational state < 45 mins. **Fail:** Plaintext PII leak or RTO > 1h.	Ensures regulatory compliance, user data confidentiality, and guarantees platform operational recovery.